

	Earth.
2ai	Normal contact (reaction) force towards centre of circular motion Weight downwards Friction upwards
ii	By Newton's 2 nd Law, (taking upwards +ve) $\sum F_y = ma = 0$ $f - W = 0$ $f = mg \quad (1)$

	Normal contact force provides the centripetal force for the passenger to be in circular motion. $\sum F_c = ma_c$ $N = \frac{mv^2}{R}$ $\frac{f}{\mu} = \frac{mv^2}{R} \quad (2)$ Taking (2) / (1): $\frac{f}{\mu} = \frac{mv^2}{R}$ $\frac{f}{mg} = \frac{v^2}{Rg}$ $v = \sqrt{\frac{Rg}{\mu}}$ where velocity v is independent of mass m
bi	$\sum F_c = ma_c$ $W - N = \frac{mv^2}{r}$ At maximum speed, $N = 0$ $W = \frac{mv^2}{r}$ $mg = \frac{mv^2}{r}$ $v = \sqrt{rg}$ $= \sqrt{95 \times 9.81}$ $= 30.528$ $= 30.5 \text{ ms}^{-1}$
ii	By Principle of Conservation of Energy, $GPE_{\text{top of hill}} + KE_{\text{top of hill}} = GPE_{\text{At A}} + KE_{\text{At A}}$ $mgh + \frac{1}{2}m(25)^2 = mg(95 \times 2) + \frac{1}{2}m(30.528)^2$ $h = 206 \text{ m}$
iii	For a larger speed, the centripetal force <u>required</u> is larger, Which means that another physical force (other than the weight) needs to act <u>towards the centre of circular path</u> which can now be provided by the track on cart.

